

PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
DARES SALAAM REGION
FORM SIX MOCK EXAMINATION
PHYSICS 3A

Code 131/3A

Time: 3:20 Hours Wednesday 14th February, 2024 A.M

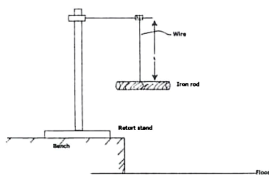
INSTRUCTIONS

1. This paper consist of **three (3)** questions.
2. Answer **all** questions.
3. All calculations must be clearly shown.
4. mathematical tables and non programmable calculator may be used.
5. Cellular phones are **NOT** allowed in the examination room
6. Write your **Examination Number** on every page of your answer booklet(s)
7. The following information may be useful
 $\text{Pie } \pi = 3.14$
Acceleration due gravity $g = 9.81 \text{ m/s}^2$
1 determine = small division on scale.

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This paper consists of 3 printed pages

1. The aim of this experiment is to determine the coefficient of rigidity of material wire given.



Proceed as follow

- Arrange the apparatus as shown in figure 1 above.
- Tie the rod on the end of the wire, making sure that the rod will hang horizontally.
- Suspend the rod on the wire from the retort stand so that the length $L = 90\text{cm}$ is used. Make the wire straight as possible.
- Start the rod oscillation in a horizontal plane (this is twisting movement, NOT swinging movement).
- Record time " t " for 20 oscillations and hence find the period T for this length.
- Repeat procedures in (c), (d) and (e) above for $L = 80\text{cm}, 70\text{cm}, 60\text{cm}, 50\text{cm}$ and 40cm .

Questions

- Tabulate your results for L (m), T (s) and T^2 (s^2)
- Plot a graph of T^2 against L
- Measure and record the radius " a " of the wire provided using a micrometer screw gauge and the mass " M " of the material rod and length (L) of the material rod (Mass in kg and length in meters)

(iv) You are given that $T = 2\pi \sqrt{\frac{2IL}{\pi\eta a^4}}$.

Where I is the moment of inertia of the rod ($I = \frac{ML^2}{12}$) and η is the coefficient of rigidity of the wire.

Find η from your graph

- (v) List two possible source of errors.

2. You are provided with beaker, thermometer measuring cylinder, some boiling water labelled W, stop watch, stirrer and paper fan.

Proceeds as follows:

- Record the room temperature θ_0 :
- Pour 100ml of boiling water into beakers, place the thermometer in a beaker containing hot water, stir gently until the temperature drops to 75°C . Stop stirring immediately start the stopwatch at this temperature and record temperature θ_1 of water at interval of 2 minutes until you get a total of seven (7) readings.
- Measure another 100ml of boiling water pour it into beaker, stir gently until the temperature drops to 75°C as in first experiment. Stop stirring immediately start the stop watch while fanning using the paper fan provided through out the experiment. Measure and record temperature θ_2 of hot water at interval of 2 minutes until you get a total of seven (7) reading

Questions

- (i) Tabulate your results for both experiment as follows

Time t (min)									
θ_1 $^\circ\text{C}$									
θ_2 $^\circ\text{C}$									

- On the same axes, plot the graph of temperature against time.
- Determine the slope S_1 and S_2 for curve 1 and curve 2 respectively at 60°C
- Calculate:

(a) The constant $K_1 = \frac{S_1}{(60 - \theta_0)^\circ\text{C}}$

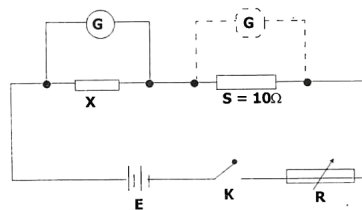
(b) The constant K_2 given that $K_2 = K_1 \left(\frac{S_1}{S_2}\right) \cdot (60 - \theta_0) \frac{1}{4}$

- Determine the ratio K_1 to K_2 , comment on your result.
- What was the aim of the experiment?

3. In this experiment you are required to determine the value of the unknown resistance X.

Proceed as follows:

Arrange the circuit as shown below, where K is the switch, R is a variable resistor, S is a standard resistor of 10Ω and X is unknown resistor.



- With $R = 400\Omega$, connect the galvanometer G first in parallel with X and record the deflection θ_x .
Remove and connect the galvanometer in parallel with S and record the deflection θ_s .
- Repeat part (a) above by increasing the value of R in the interval of 100Ω . Each time record the deflections so as to obtain a total of seven readings.

Questions

- (i) Tabulate your results for both experiment as follows

R (Ω)	400	500	600	700	800	900	1000
θ_x							
θ_2							

- Plot the graph of deflection θ_x against θ_s .
- From your graph obtain the value of a slope.
- Use the slope obtained in (iii) above to find the value of X
- State any two source of errors.

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FORM VI PHYSICS 3A

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1 (A) (i) Table of results for metal rod.

L in cm	t in seconds	Period T (s)	T ² (seconds) ²
90	183.6	9.18	84.27
80	151.8	7.59	57.61
70	147.6	7.38	54.46
60	140.4	7.02	49.28
50	132.6	6.63	43.96
40	123.6	6.18	38.19

(06 marks)
@ column
or

(ii) The graph of T²(s)² against L(m) is in the graph ^{paper} behind.

(iii) Diameter of the wire d = 0.34 mm that is radius

$$a = 0.17 \text{ mm}$$

the mass M of the rod = 126 g (its range 125g - 130g)
= 0.126 kg

the metal rod l = 15 cm

$$l = 0.15 \text{ m}$$

(01 1/2 marks)
(0.017 - dia, 0.017 - radius
0.126 - mass)

(iv) Given $T = 2\pi \sqrt{\frac{2IL}{\pi \eta a^4}}$

Where $I = \frac{1}{12} ML^2$ (moment of inertia)

η = the coefficient of rigidity of the wire.

From the graph slope $S = \frac{\Delta T^2}{\Delta L}$ (00 1/2 marks)

$$= \frac{49 - 12}{0.6 - 0.15} = 82.25 \text{ s}^2/\text{m} \text{ (00 1/2 marks)}$$

then $T^2 = 4\pi^2 \left(\frac{2IL}{\pi \eta a^4} \right)$ Range (71.15 s²/m — 142.25 s²/m)

$$\frac{T^2}{L} = \frac{8\pi I}{\eta a^4}$$

$$\text{Slope } S = \frac{8\pi I}{\eta a^4} \quad \leftarrow \left(\frac{1}{2} \text{ marks} \right)$$

$$\eta = \frac{8\pi I}{a^4 S} \quad \leftarrow \left(\frac{1}{2} \text{ marks} \right)$$

$$= \frac{8\pi}{a^4 S} \left(\frac{1}{12} ML^2 \right) = \frac{2\pi ML^2}{3a^4 S} \quad \leftarrow (0.1 \text{ marks})$$

$$= \frac{2\pi \times 0.126 \times (15)^2}{82.2 \times (0.17 \times 10^{-3})^4}$$

$$\eta = 0.86 \times 10^{11} \text{ Pa} \quad \leftarrow \left(\frac{1}{2} \text{ marks} \right)$$

Range of coefficient of rigidity $(0.5 \times 10^{11} \text{ — } 1 \times 10^{11}) \text{ Pa}$.

(V) Sources of error - Any two.

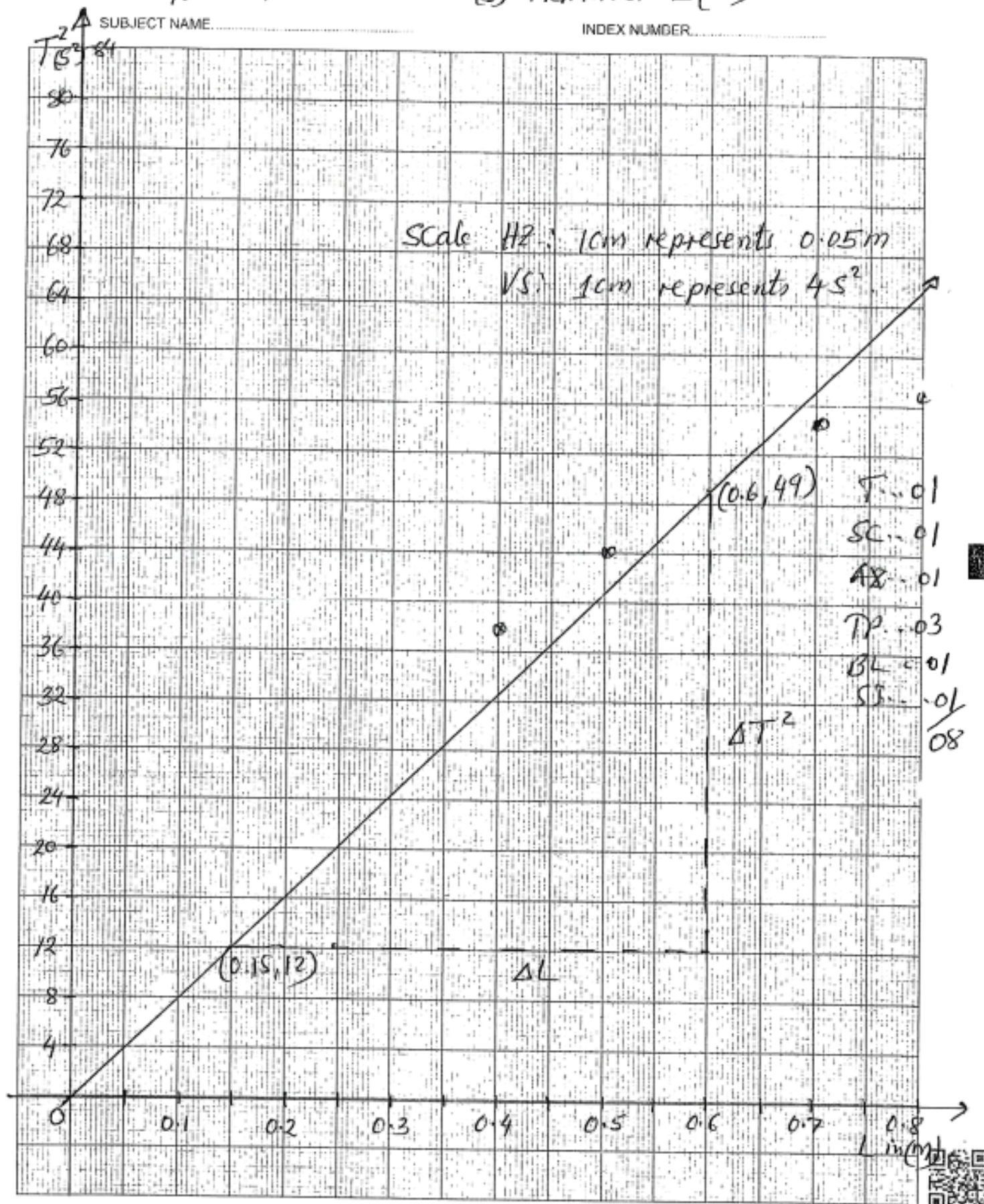
→ Parallax error

→ Zero error

→ Time reaction (due to starting and ending)

→

THE GRAPH OF $T^2(s)^2$ AGAINST $L(m)$.



2d: Table of Result (Room temperature $\theta_0 = 30^\circ$)

Time t (in min)	0	2	4	6	8	10	12	14
θ_1 $^\circ\text{C}$	75	69	65	61	58	56	54	52
θ_2 $^\circ\text{C}$	75	66	59	54	50	47	44	41

(4 marks)
or 2 @ row

(f) The slopes:

Slope in $\theta_1 = S_1 = \frac{\Delta\theta_1}{\Delta t_1}$ (or $\frac{1}{2}$ marks) and $S_2 = \frac{\Delta\theta_2}{\Delta t_2}$ (or $\frac{1}{2}$ marks)

$$S_1 = \frac{62 - 56.4}{5.5 - 9.3}$$

$$S_2 = \frac{64 - 52}{2.5 - 6}$$

$$= -1.47^\circ/\text{min} \text{ (or } \frac{1}{2} \text{ marks)}$$

$$= -3.43^\circ/\text{min} \text{ (or } \frac{1}{2} \text{ marks)}$$

(g) (i) From $k_1 = \frac{S_1}{(60 - \theta_0)}$

(ii) $k_2 = k_1 \left(\frac{S_1}{S_2} \right) \times (60 - \theta_0)^{\frac{1}{4}}$

$$k_1 = \frac{-1.47}{(60 - 30)}$$

$$= -0.049 \times \left(\frac{-1.47}{-3.43} \right) (60 - 30)^{\frac{1}{4}}$$

$$= -0.049/\text{min} \text{ (or } \frac{1}{2} \text{ marks)} = -0.049^\circ/\text{min} \text{ (or } \frac{1}{2} \text{ marks)}$$

(ii) The ratio of k_1 to $k_2 = \frac{k_1}{k_2}$

$$= \frac{-0.049}{-0.049} = 1:1. \text{ (or 1 mark)}$$

(h) The aim of the experiment is to compare the rate of cooling of a liquid in forced and natural convection. (or 1 mark)

Comment

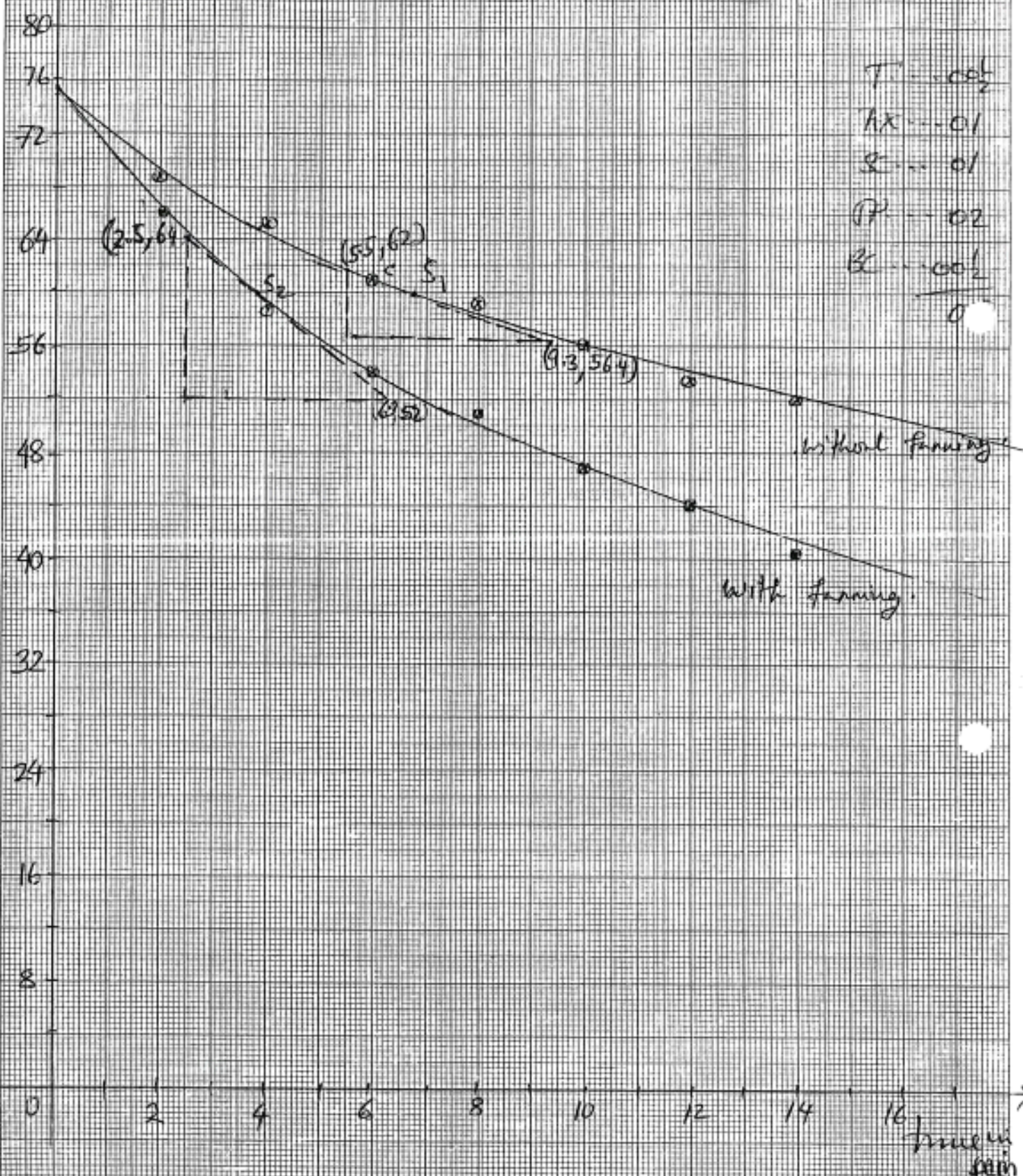
- The rate of cooling under forced condition is greater than under natural condition since $|S_2| > |S_1|$. (or 1 mark)

- But $k_1 = k_2$, since both experiment same material is used (water) then convection constant is the same.

Q.11A THE GRAPHS OF TEMPERATURE AGAINST TIME IN MIN

Scale H.B: 1cm represents 15sec

V.S: 1cm represents 4°C.



Q3. (a) Table of Results.

R (Ω)	400	500	600	700	800	900	1000	
D_x	5	4	3	2.5	2	1.5	1.5	
D_s	8	7	6	5	4	3.5	3.5	

(ii) Graph on graph paper. (0.4 miles)
OR 3 rows.

(iii) From, $\text{slope} = \frac{\Delta D_x}{\Delta D_s}$ (0.01 miles)

$$\text{slope} = \frac{3-1}{6-2}$$

$$\text{slope} = 0.5$$

(0.01 miles)
(Range of slope 0.45-0.54)

(iv) The value of x .

$$V \propto D \quad \text{(0.01 miles)}$$

$$V = cD$$

$$IR = cD$$

Then, $I_x R_x = cD_x$ and $I_s R_s = cD_s$.

$$\frac{I_x R_x}{I_s R_s} = \frac{cD_x}{cD_s} \quad \text{(0.1 miles)}$$

$$\text{Since } I_x = I_s$$

$$\frac{R_x}{R_s} = \frac{D_x}{D_s} \quad \text{(0.01 miles)}$$

$$R_x = \left(\frac{D_x}{D_s} \right) R_s \quad \text{but slope} = \frac{D_x}{D_s}$$

$$R_x = \text{slope} \times R_s$$

$$R_x = 0.5 \times 10 \quad \text{(0.1 miles)}$$

$$R_x = 5 \Omega \quad \text{(Range of } R_x \text{ 4.5 - 5.4) } \Omega$$

(v) Source of errors (any two)

- Parallax error

- Zero errors

-

(0.1 miles) @ 0.01 miles

THE GRAPH OF DEFLECTION Δ_x AGAINST Δ_s

Δ_x

5.0

4.5

4.0

3.5

3.0

2.5

2.0

1.5

1.0

0.5

0

1

2

3

4

5

6

7

8

9

Scale: H.S 1cm represents 0.5 deflection

V.S 1cm represents 0.25 deflection

(6, 3)

(2, 1)

Δ_s

Δ_x

T. = 0.05

SC = 0.1

AP = 0.1

TP = 0.1

BL = 0.05

SI = 0.05

06